Quant-Enabled Healthcare Resource Allocation: Optimization For Underrepresented Communities

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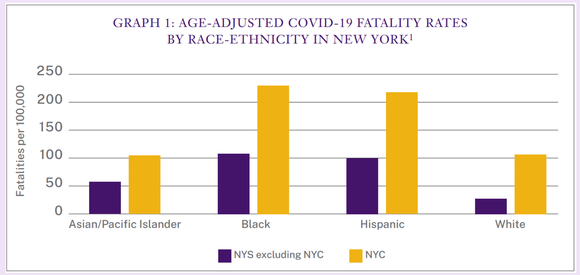
Introduction

Healthcare disparities continue to impact underrepresented communities, leading to unequal access to medical resources and worse health outcomes. During events like the Covid-19 pandemic, black and Hispanic new Yorkers experienced a much higher fatality rate which was more than double compared to their white counterparts.

These disparities are driven by socioeconomic factors such as high exposure risk among essential workers, limited healthcare access, and housing density.

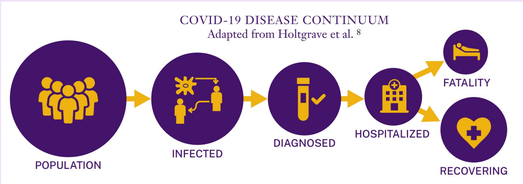
Quantum-AI presents a revolutionary solution. This research applies the Variational Quantum Eigensolver (VQE) and IBM's Qiskit platform to optimize healthcare logistics, enabling faster, more efficient, and data-driven resource allocation. By integrating Quantum Algorithms with classical ai processing we aim to optimize this resource allocation leading to minimized disparities in underrepresented communities.

During the COVID-19 pandemic, hospitals in underrepresented communities faced severe shortages of ICU beds, ventilators, and healthcare workers, leading to disproportionately higher mortality rates. Traditional resource allocation models failed to account for the unique socio-economic factors affecting these communities



"COVID-19 fatality rates for Black and Hispanic New Yorkers were more than twice as high as for White residents, revealing critical disparities in healthcare access and resource distribution"

University at Albany, SUNY, NYS COVID-19 Minority Health Disparities Team. *Understanding and Eliminating Minority Health Disparities in a 21st-Century Pandemic: Issue Brief #1*. July 2020, http://www.albany.edu/mhd.



The COVID-19 Disease Continuum traces infection to potential outcomes, highlighting disparities in healthcare access that worsen inequities for underrepresented communities. Quantum AI-driven optimization can improve resource distribution, ensuring fairer and more effective healthcare allocation.

Holtgrave, D. R., et al. *Assessing racial and ethnic disparities using a COVID-19 outcomes continuum for New York State.* Annals of Epidemiology, 2020. DOI:10.1016/j.annepidem.2020.06.010.

Materials

Data & Computing Tools:
- **CDC COVID-19 Data Tracker** – Real-world pandemic data.
- **AI PC Development Kit** – High-performance AI computing.

Quantum & AI Software:
- **Qiskit** – IBM's quantum computing framework.
- **Variational Quantum Eigensolver (VQE)** – Optimizing resource allocation.
- **Cost Function Optimization** – Minimizing healthcare disparities.

Hardware Specs:
- **Intel® Core™ Ultra 7 Processor 155H**
- **Compact Form Factor:** 4.6" x 4.4" x 1.5"
- **Memory & Storage:** 32GB RAM, 512GB SSD

Methodology

Solution with Variational Quantum Eigensolver (VQE):

- **Define decision variables** – Optimize allocation of critical medical resources.
- **Objective function** – Maximize resource utilization, minimize wait times, and balance supply distribution.

VQE Implementation:

- **Map the allocation problem** to a Hamiltonian system.
- **Define constraints and objective function** to align with healthcare needs.

Qiskit Design & Execution:

- **Construct the Hamiltonian** representing the optimization problem.
- **Design quantum circuits** for VQE-based optimization.
- **Integrate classical computing** with quantum simulations to refine allocation strategies.

The Quantum Algorithm's Role

The Variational Quantum eigen solver (VQE) will adjust the decision variables $X_{i,j}$, $Y_{i,j}$, $Z_{i,j}$, dynamically to minimize the total cost:

$$C(X,Y,Z) = \alpha C_1 + \beta C_2 + \gamma C_3$$

- This function indicates that if patient wait times are high, then the VQE allocates more ICU beds/staff
- If a hospital is underutilized, the VQE shifts resources elsewhere
- If there is demand-supply mismatch, the VQE will optimize and rebalance allocations

*"The **Variational Quantum Eigensolver (VQE)** optimizes healthcare resource distribution by dynamically adjusting decision variables to minimize a predefined **cost function**. The formula represents a **weighted optimization model**, where different cost components account for factors like **hospital resource utilization, patient wait times, and demand-supply balance**."*

Results

What's the goal ?

The goal of this project is to find the best way to distribute **medical resources** like **ICU beds**, **ventilators**, and **personal protective equipment (PPE)** across hospitals in Brooklyn, New York, a city severely affected by 2020s COVID-19 Pandemic. We use the VQE to make smarter decisions about where resources will go.

Key Points

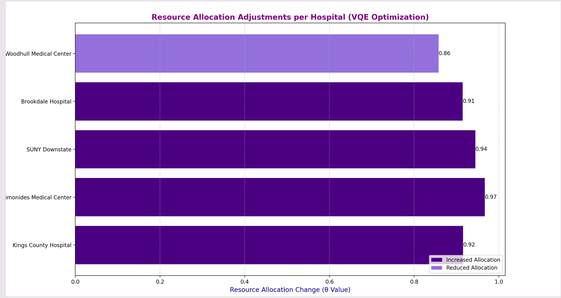
- Hospitals with higher patient demand got more resources
- Hospitals with lower demand wastes less resources
- Resource allocation normalized between **0** (least allocation) and **1** (most allocation) for each category

Findings

1. ***Algorithm Performance***
   - Final score of the **VQE** is -2.57717
   - The lower the score the better the resource distribution
2. ***How were resources distributed***
   - **Five** major hospitals were chosen
   - Resources adjusted based on hospital's capacity and needs using **real** data for ICU beds, ventilators, and PPE.

	Hospital	ICU Beds	...	Normalized Ventilators	Normalized PPE
0	Kings County Hospital	80	...	0.857143	1.00
0	Kings County Hospital	80	...	0.857143	1.00
1	Maimonides Medical Center	90	...	1.000000	0.90
1	Maimonides Medical Center	90	...	1.000000	0.90
2	SUNY Downstate	85	...	0.928571	0.96
2	SUNY Downstate	85	...	0.928571	0.96
3	Brookdale Hospital	82	...	0.892857	0.94
4	Woodhull Medical Center	78	...	0.785714	0.92

What does the graph show ?



Dark Purple = received more resources

Light Purple = received fewer resources

- **Maimonides received most ventilators**
- **Kings Country received most PPE**
- **Woodhull got fewer everything due to its size**

Conclusion & Achievements

- The algorithm minimized inefficiencies by distributing resources based on hospitals real world capacity & demand
- Achieved an optimized cost function value of -2.57717, indicating a more balanced and effective strategy
- Incorporated real hospital data from Brooklyn's major healthcare centers during critical events such as the COVID-19 Pandemic
- Normalization allowed model to treat all hospitals fairly, regardless of the size or capacity.

Implications for healthcare management

- This quantum framework can help policy makers and healthcare system respond more effectively during crisis's.

Conclusion Statement

- This study highlights the potential of quantum algorithms like VQE in transforming healthcare resource management. By using real-world data and advanced optimization techniques, we can create a more **equitable**, **efficient**, and **responsive** healthcare system—especially for communities historically impacted by health disparities.

Recommendations

- **Adopt Quantum AI in Healthcare** – Implement **VQE-based optimization** for efficient resource allocation.
- **Utilize Real-Time Data** – Leverage AI to dynamically adjust hospital resource distribution.
- **Expand Research on Quantum Scalability** – Investigate how quantum computing can optimize **nationwide healthcare systems**.
- **Enhance Security & Accessibility** – Ensure **data privacy in quantum AI models** and increase access for researchers.

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